External road checks for western Democratic Republic of the Congo

This supplement reports a completed comparison of mapped road inputs with external satellite detections and a historical traffic-bearing corridor inventory. It documents substantial disagreement for newly developed forest roads. It does not validate population coverage, travel behaviour, or the feasibility of new urban hubs.

Sources and scope

The OSM ZIP hash exactly matches the October 2026 road extract used in the revised paper. Study masking uses the same ten provinces as the main analysis. Slagter et al.'s version 1.04 archive MD5 matches the producer checksum 15af527668b495187c25b2964188ac69. The reference describes monthly road development detected from 10-m Sentinel-1/2 imagery in 2019–2025; it excludes older roads not detected as developments. Its own model validation is not validation of our study. The 2024 paper reports a detection assessment for an earlier period, which is not assumed to establish accuracy of every 2025 reference line. No raw forest-road archive is redistributed here because the record does not show an unambiguous licence in the retrieved metadata.

Spatial comparison method

An external input check compares the October 2026 OSM extract with Slagter et al.'s satellite-detected 2019–2025 forest-road development (Slagter et al., dataset and paper). The reference is clipped to the ten-province union. Lines are projected to a local azimuthal equidistant CRS (19°E, 1°S), divided into subsegments no longer than 25 m, and sampled at subsegment midpoints. Each sample receives its subsegment length as weight. Agreement at tolerance t is the length-weighted fraction whose nearest mapped-road line lies within t. Tolerances are 25, 50, 100, and 250 m; OSM road sets are the main classes, main classes plus tracks, and all mapped line classes. Repeating the calculation at 50 m checks numerical sampling sensitivity. Road geometry agreement does not test graph junctions, traversability, travel time, or coverage. The satellite product represents newly detected forest roads rather than the entire public road network, has its own detection errors, and is not treated as field ground truth. Possible subsequent OSM reuse of satellite mapping cannot be ruled out.

For each line of projected length L, n = ceil(L/25 m) equally spaced subsegment midpoints receive weight L/n. Agreement $A(t) = 100 \times \Sigma\, w_i\, 1[d_i \le t] / \Sigma\, w_i$, where d_i is the nearest-line distance. This approximates matched line length, not counts of matched roads. Nearest-line matching is one-sided and can match unrelated nearby parallel lines; it does not estimate precision. Four buffers were retained rather than selecting the most favourable one after inspection. The local projection is an approximation to surface distances, and border clipping may affect the small subset near the study boundary. Midpoints are not independent observations, so no standard-error or confidence-interval calculation is assigned to the sample count.

Satellite road agreement

The clipped satellite reference contains 12,798 line fragments totalling 2,062.53 km (Table 1). At a 100-m tolerance, length-weighted agreement is 4.91% for the main-road set, 20.93% with tracks, and 20.96% for all mapped classes. At 250 m, these shares are 5.58%, 24.43%, and 24.52%. Across both reference inventories and all road-set/tolerance combinations, changing midpoint sampling from 25 to 50 m changes agreement by at most 0.113 percentage points. Low agreement provides an external warning about representation of recently developed forest roads; it cannot be interpreted as a national OSM completeness estimate. Tracks dominate the additional matches. Detected logging roads may be private, seasonal, abandoned, or unsuitable for public access, so unmatched lines are not inserted into the operational graph automatically. Their unmatched fraction is not substituted for the model's assumed 1% random-deletion rate.

TABLE 1
LENGTH WEIGHTED AGREEMENT WITH SATELLITE DETECTIONS

Road set	25 m	50 m	100 m	250 m
Main classes	4.19	4.63	4.91	5.58
Including tracks	16.41	18.68	20.93	24.43
All mapped classes	16.42	18.70	20.96	24.52

The 25-m run uses 88,832 weighted reference midpoints. The 50-m check uses 48,416. Both represent the same 2,062.53-km clipped reference. They are numerical quadrature samples, not journeys, traffic stations, or independent road surveys.

Historical traffic inventory

The World Bank AICD road file supplies 859 records (World Bank AICD). After removing two zero-ID dummy records and excluding the −99 missing sentinel, only five segments have positive AADT values: one value of 510 and four repeated values of 966. Documentation dates traffic to 2004, describes units as passenger-vehicle units, and warns that only one road has traffic information. These are not five independent count stations. The five segments total 323.36 km of projected reference geometry; main-road spatial agreement is 84.30% at 100 m and 92.55% at 250 m. Because this historical catalogue may share source cartography with OSM, this is a corridor-consistency benchmark, not an independent predictive test. The study does not predict AADT, so neither traffic value is used to calculate model error or validate population coverage.

The national file contains 852 missing traffic records encoded −99 and two zero-ID dummy records. The four records carrying 966 repeat one link name across administrative subdivisions, while their START_NODE/END_NODE labels are inconsistent with the full link name. No origin–destination route benchmark is inferred from those labels. Summing traffic over these records would double-count a corridor value. The published metadata explicitly describes poor traffic-data coverage, so the positive values are documented without extrapolating a regional flow model.

What this adds and what remains

Completed evidence: downloaded external references, checksum verification, reproducible masked spatial comparison, and a sampling-resolution check. The results justify concern about spatially structured map omissions and distinguish exclusions of tracks from unmapped detections. They do not calibrate an omission distribution for all transport classes, confirm current traversability, or supply a time-speed error distribution. Random deletion at 1% remains a declared stress assumption. Simply replacing it with 79.07% would be invalid because the satellite sample is restricted to recent forest-road development.

For the main covering objective, independent observed access or journey records must be matched to the same origins, destinations, and network scope. A secondary dataset can satisfy this without a new field campaign if it contains compatible geocoded measurements. Expert interviews concern normative weights or planning feasibility; they cannot replace those measurements. The main covering objective itself contains no elicited multi-criteria weights.

Lecture pratique en français

Nous avons maintenant un contrôle externe effectivement calculé sur la cartographie routière. Il révèle un problème, au lieu de confirmer artificiellement le modèle : une grande partie des routes forestières récentes détectées par satellite est absente du réseau cartographié, même lorsque les pistes sont incluses. Ce contrôle renforce la réponse sur la qualité des données. Il ne permet pas encore de dire que les déplacements des habitants ou la couverture de population ont été validés. Les deux anciennes valeurs de trafic ne sont pas une base régionale suffisante. Il reste donc à obtenir une mesure compatible de trajet ou d’accès pour valider ce résultat précis.

References

[33] B. Slagter et al., “Forest roads (Congo Basin),” ver. 1.04, Zenodo, doi: 10.5281/zenodo.20201702. Dataset file: forestroads_afr_2019-01_2025-12.zip. Accessed: 7 Oct. 2026.

[34] B. Slagter et al., “Monitoring road development in Congo Basin forests with multi-sensor satellite imagery and deep learning,” Remote Sensing of Environment, vol. 315, art. 114380, 2024, doi: 10.1016/j.rse.2024.114380.

[35] World Bank, “Congo, Dem. Rep. Roads,” AICD/Tecsult road inventory and documentation. [Online]. Available: https://datacatalog.worldbank.org/search/dataset/0040251/congo-dem-rep-roads. Accessed: 7 Oct. 2026.